Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 2.0, **Version Date** 2/24/13

Beam on Window (BOW) Alignment Process

(Click Here for Summary Version)

1 Personnel Requirements

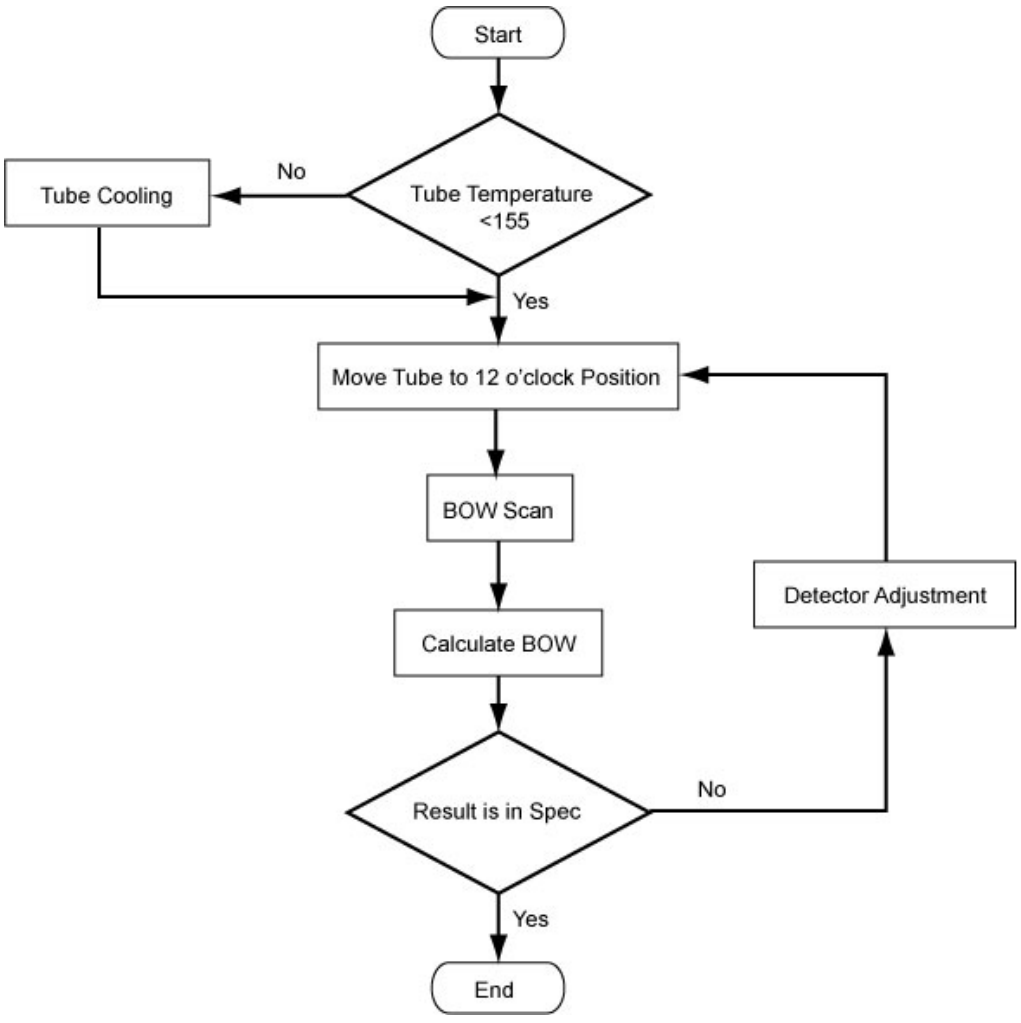
Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

2 Overview

The objective of this procedure is to put the detector in the correct position, assuming the tube is already in the correct position. This makes sure the X-ray Flux does not miss the detector.

NOTE: When performing alignments, wait at least 15-30 minutes between scans to prevent unnecessary adjustments.

Illustration 1: BOW Flowchart



3 Preliminary Requirements

3.1 Safety

⚠ CAUTION
No personnel or patients should be in the area of the gantry while scanning without adequate radiation safety precautions being utilized.

NOTICE
Follow ALL required safety and PPE procedures customary for your organization, when working on this product.

3.2 Required Conditions

Condition	Reference	Effectivity
Recommend procedure completion by GE Service trained personnel.	-	-
The positioning lights must be OK before BOW alignment.	-	-

4 Procedure

4.1 For Tube Change Only

Wait 90 minutes if the new tube had more than 25 Kilo Joules of energy input, [KV x mA x Sec ÷ 1000] within the last 30 minutes prior to the start of system alignments. If a tube heat soak has been performed you must wait a minimum of 6 hours before system alignments can be performed.

4.2 Confirm Tube Cooling

When performing alignments, wait at least 15-30 minutes between scans to prevent unnecessary adjustments.

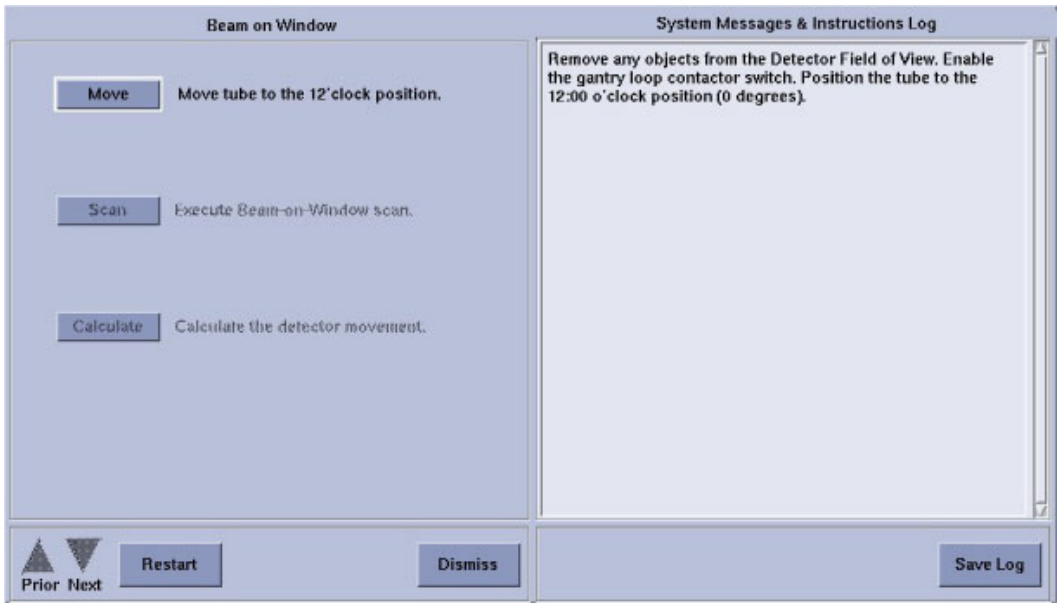
4.3 Preparation

1. Move the cradle OUT to its fully retracted position (farthest from gantry) and position the table to its lowest height.
2. Remove the gantry left side cover.
3. Turn OFF “Axial Enable”, “HVDC” and “120VAC” switches on the service switch panel.
4. Remove the table foot switch cover.
5. Remove the gantry covers as required.
6. Turn ON “HVDC”, “Axial Enable” and “120VAC” switches.

4.4 Adjustment Procedure

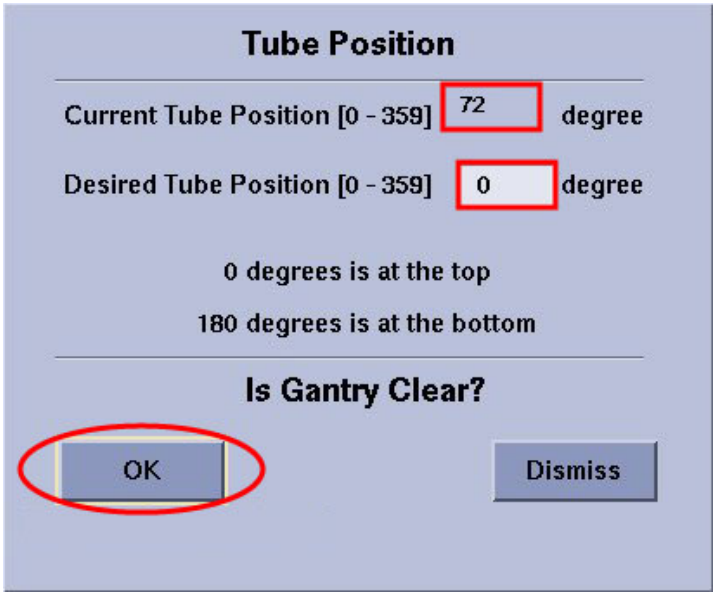
1. Select [Service] ->[Calibration] -> [BOW Alignment] to enter ‘Beam on Window’ UI.

Illustration 2: Beam on Window



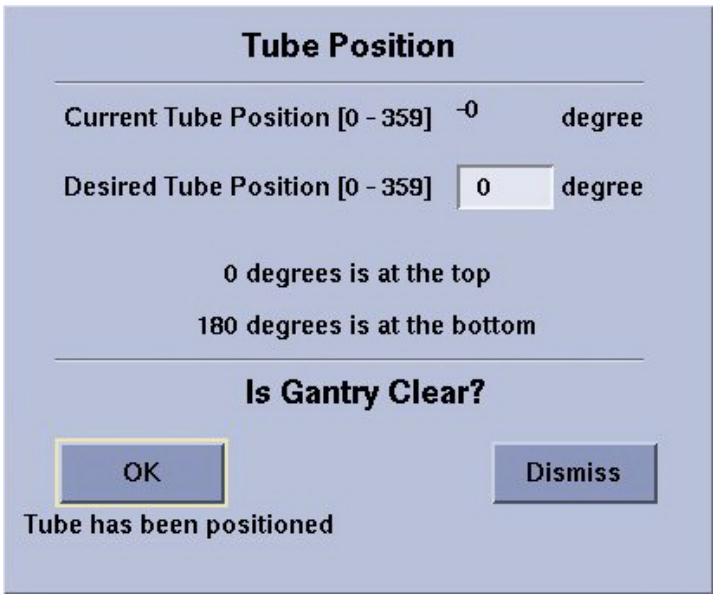
2. Remove any objects from the detector field of view.
3. Click [Move] to enter ‘Tube Position’ window.
    a. Check ‘Current Tube Position’, if not 0 deg, input **0** in ‘Desired Tube Position’. Then click OK.

Illustration 3: Tube Position



b. After 'Current Tube Position' displays 0 degree, click Dismiss to quit this window.

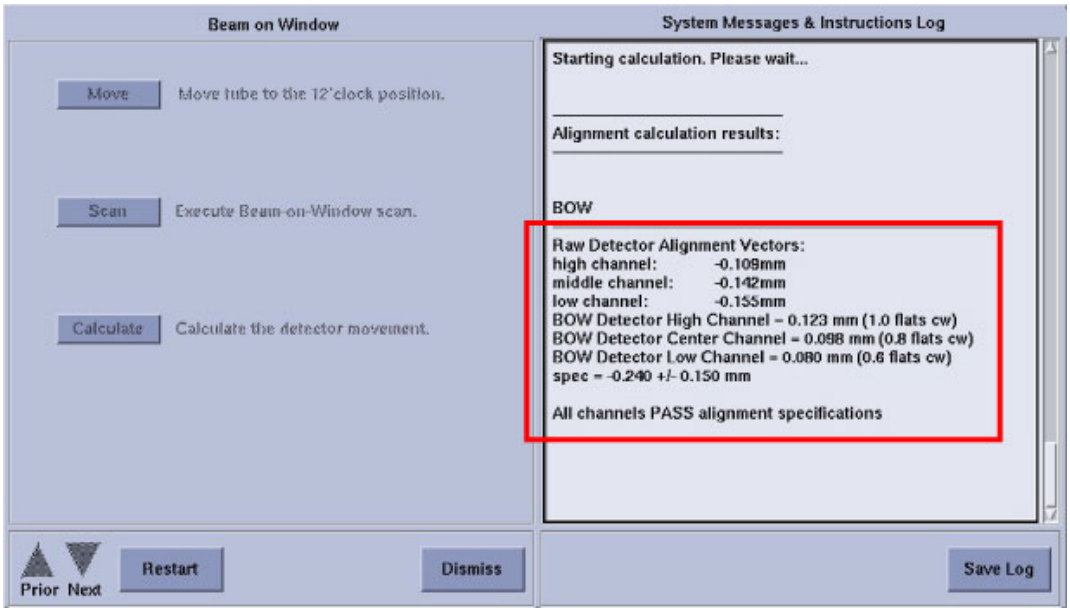
Illustration 4: Tube Position OK



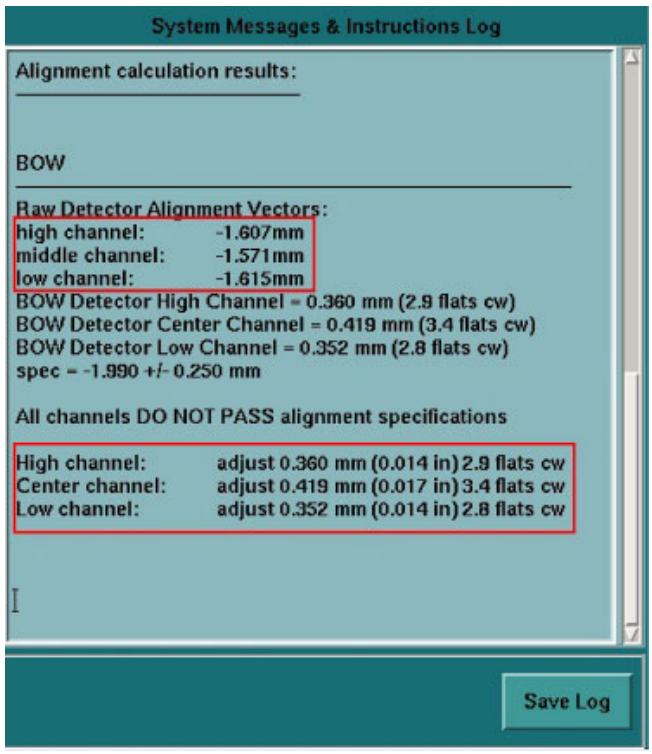
4. Click [Scan] to execute Beam-on-Window scan.
5. Click [Calculate] to calculate the detector movement, and the account result displays on the monitor.

   Verify values are within specification -0.240mm ± 0.150mm.

   - If BOW is within spec, DO NOT adjust anything.

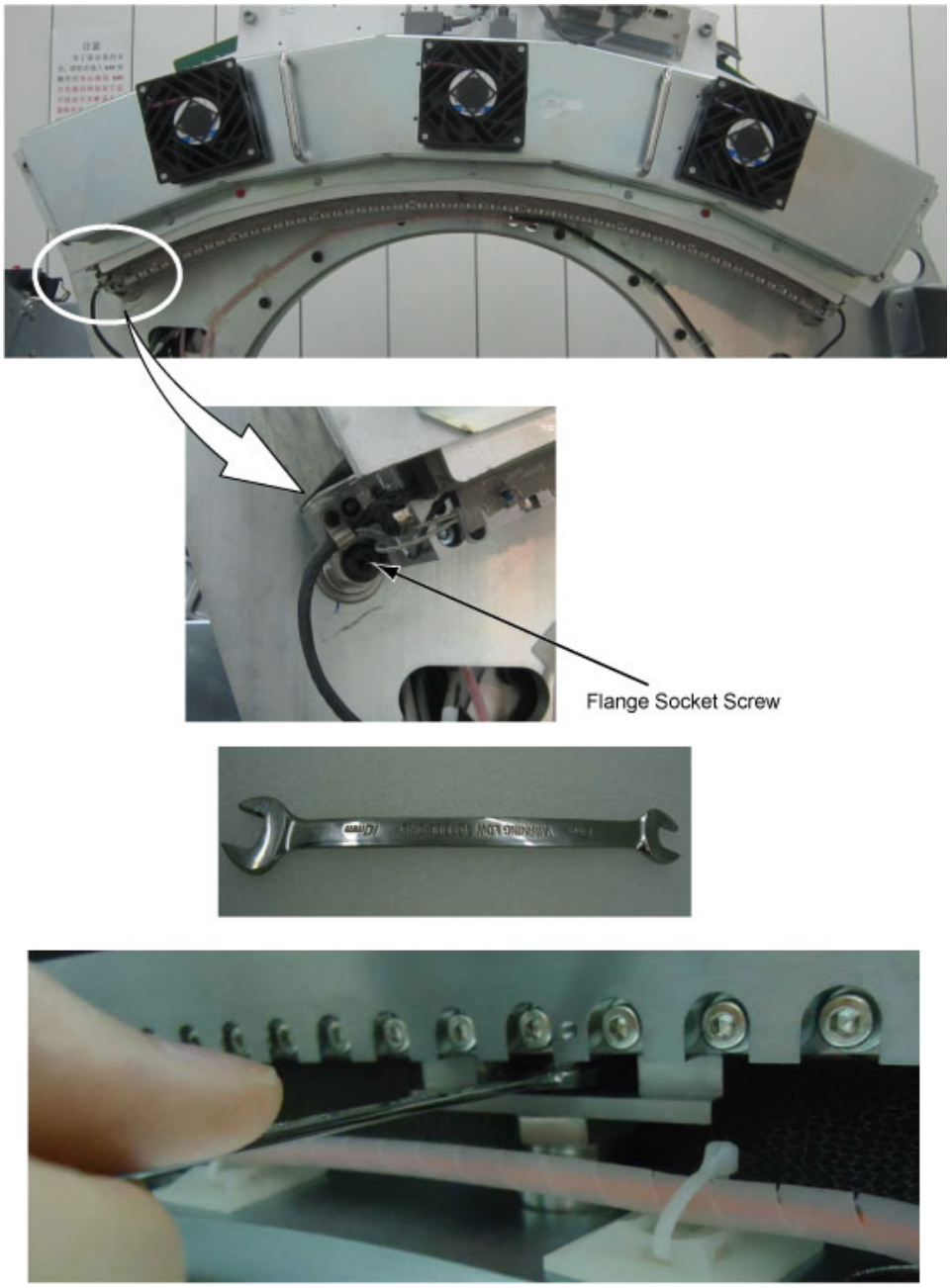


- If BOW is out of spec, continue the following steps to execute the detector adjustment.



6. Adjust the detector position.
   a. Turn OFF “HVDC”, “Axial Enable” and “120VAC” switches on the service switch panel.
   b. Loosen two flange socket screws in both sides of the detector and the lock screw in the middle of the detector.

Illustration 5: Flange Socket Screws and Lock Screw



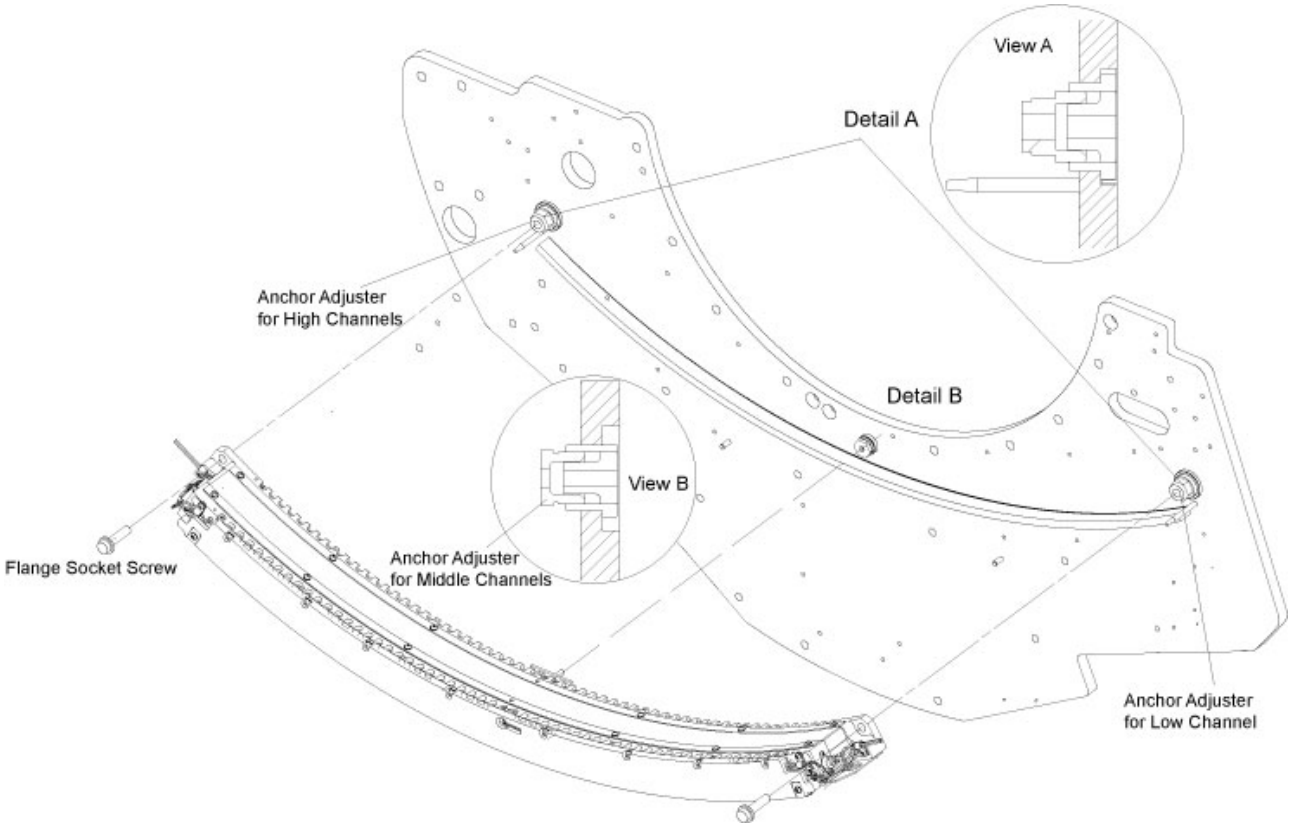
c. Turn three anchor adjusters (See Illustration 6) to shift the detector offset within specification along the Z-Axis according to the calculative result displayed on the monitor screen.

NOTE:

One turn  = 6 Flats

- Clockwise: BWD, away from the table
- Counterclockwise: FWD, towards the table

Illustration 6: Beam on Window Adjustment



7. Secure the lock screw with torque 13.5N-m and two flange socket screws with torque 30.5N-m which were loosened in Step 6.b.
8. Turn ON “Axial Enable”, “HVDC” and “120VAC” switches.
9. Click [Restart] to repeat BOW procedure from Step 3 tdo Step 5.
10. Click [Save Log], then click [Dismiss] to quit BOW.
11. Restore the gantry to original configuration and reinstall the table foot switch cover.

5 Finalization

No finalization steps.

Back to top